1．$\Sigma = \{0,1\}$，根据要求写出正则表达式

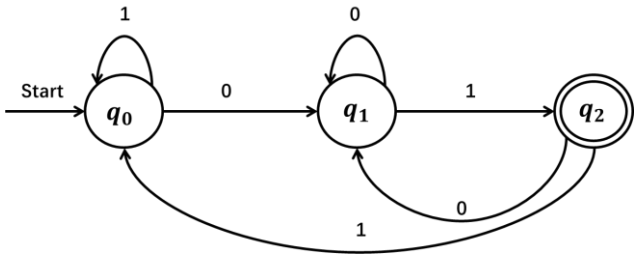
(a) $L = \{w | w \in \Sigma^*, w$中 0 和 1 的个数都是偶数$\}$

$(00 + 11 + (01 + 10)(00 + 11)^*(01 + 10))^*$

(b) $L = \{w | w \in \Sigma^*, w$不包含 101 子串$\}$

$0^*(1 + 000^*)^*0^*$ 或$(0 + \varepsilon)(1 + 000^*)^*(0 + \varepsilon)$或$(0 + \varepsilon)(1 + 00 + 000)^*(0 + \varepsilon)$

2．将下图所示的 DFA 用递归构造法造出等价的正则表达式。



（1）递归构造法：

$$R_{13}^{(1)} = R_{13}^{(0)} + R_{11}^{(0)}\left(R_{11}^{(0)}\right)^* R_{13}^{(0)} = \emptyset + \emptyset = \emptyset$$

$$R_{12}^{(1)} = R_{12}^{(0)} + R_{11}^{(0)}\left(R_{11}^{(0)}\right)^* R_{12}^{(0)} = 0 + (\varepsilon + 1)(\varepsilon + 1)^*0 = 1^*0$$

$$R_{22}^{(1)} = R_{22}^{(0)} + R_{21}^{(0)}\left(R_{11}^{(0)}\right)^* R_{12}^{(0)} = (\varepsilon + 0) + \emptyset = \varepsilon + 0$$

$$R_{23}^{(1)} = R_{23}^{(0)} + R_{21}^{(0)}(R_{11}^{(0)})^* R_{13}^{(0)} = 1 + \emptyset = 1$$

$$R_{33}^{(1)} = R_{33}^{(0)} + R_{31}^{(0)}(R_{11}^{(0)})^* R_{13}^{(0)} = \varepsilon + 1(\varepsilon + 1)^*\emptyset = \varepsilon$$

$$R_{32}^{(1)} = R_{32}^{(0)} + R_{31}^{(0)}(R_{11}^{(0)})^* R_{12}^{(0)} = 0 + 1(\varepsilon + 1)^*0 = 0 + 11^*0$$

$$R_{13}^{(2)} = R_{13}^{(1)} + R_{12}^{(1)}\left(R_{22}^{(1)}\right)^* R_{23}^{(1)} = \emptyset + (1^*0)(\varepsilon + 0)^*1 = 1^*00^*1$$

$$R_{33}^{(2)} = R_{33}^{(1)} + R_{32}^{(1)}(R_{22}^{(1)})^* R_{23}^{(1)} = \varepsilon + (0 + 11^*0)(\varepsilon + 0)^*1 = \varepsilon + (0 + 11^*0)0^*1$$

$$R_{13}^{(3)} = R_{13}^{(2)} + R_{13}^{(2)}(R_{33}^{(2)})^* R_{33}^{(2)}$$

$$= 1^*00^*1 + (1^*00^*1)(\varepsilon + (0 + 11^*0)0^*1)^*(\varepsilon + (0 + 11^*0)0^*1)$$